

Sprint Report – Sprint 3

Sprint Goal

Establish a complete data pipeline foundation for the CMOS sensor dataset, including data parsing, preprocessing, feature engineering, and initial data understanding to enable downstream modelling.

Sprint Duration

16 March – 22 March 2026

Team Roles

- Faezeh & Sepideh: Data pipeline implementation, feature engineering, notebooks
 - Adi: Scrum master
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Sprint Backlog & Progress

Completed Tasks

- [x] Task 1 — Define core data contracts
 - [x] Task 2 — Configuration loader
 - [x] Task 3 — I/O utilities
 - [x] Task 4 — Raw data parser
 - [x] Task 5 — Metadata tracking
 - [x] Task 6 — Preprocessing
 - [x] Task 7 — Signal filtering & baseline correction
 - [x] Task 8 — Exposure detection
 - [x] Task 9 — Temporal feature extraction
 - [x] Task 10 — Structured feature table
 - [x] Task 11 — End-to-end ingestion flow
 - [x] Task 12 — Minimal training pipeline stub
 - [x] Task 13 — Basic tests for pipeline
 - [x] Task 14 — Toluene integration boundary documentation
 - [x] Task 15 — Scrum documents
 - [x] Data Understanding notebooks and documentation
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Key Deliverables

1. End-to-End Data Pipeline

- Raw CMOS sensor data → structured dataset
- Modular pipeline design:
 - Parser
 - Preprocessing
 - Feature Engineering
 - Storage layer

2. Data Understanding

- Transformation of 32×32 sensor frames into 1024 features
- Statistical analysis:
 - Mean, standard deviation
 - Distribution analysis
- Visualizations:
 - Histograms
 - Heatmaps
- Temporal analysis of signal behaviour

3. Feature Engineering

- Frame-level features:
 - mean, std, max
- Temporal features:
 - differences (Δ)
 - rolling statistics
- Phase-based features:
 - baseline / exposure / recovery

4. Data Quality Insights

- Presence of many zero-valued frames → requires cleaning
- Signal saturation near maximum values (~4091)
- Temporal patterns are more informative than spatial patterns

Sprint Metrics

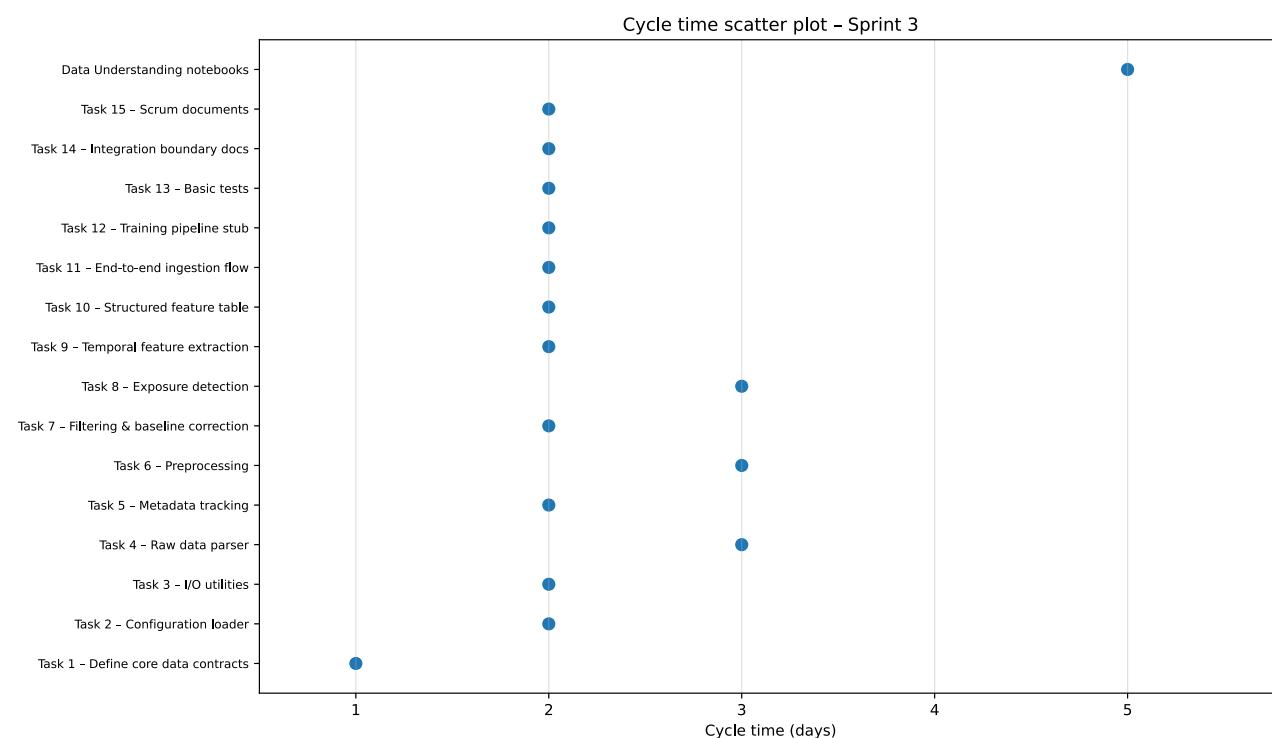
Cycle Time (Estimated)

Task Group	Duration
Core Pipeline	3–4 days
Feature Engineering	2–3 days
Integration	1–2 days
Documentation	1–2 days

Average Cycle Time: ~2.5 days
Total Tasks Completed: 16
Completion Rate: 100%

Cycle Time Analysis

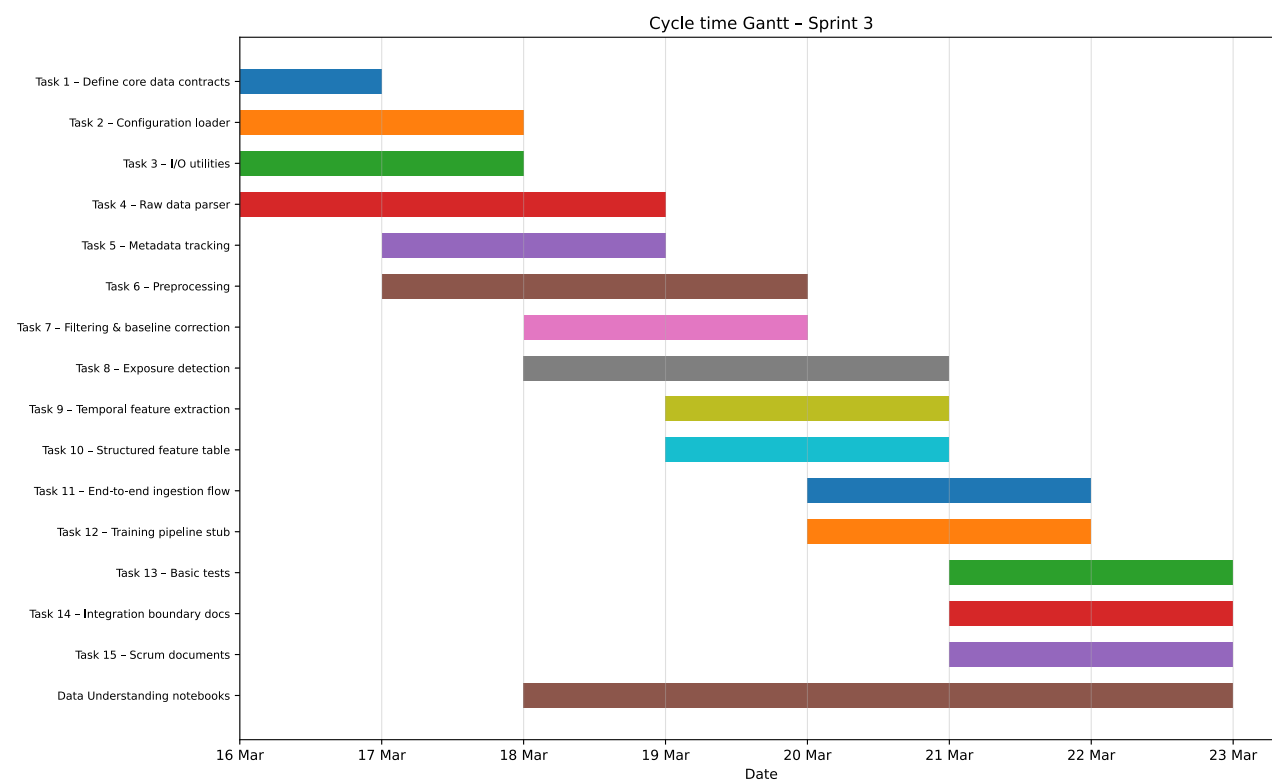
Cycle Time Scatter Plot



Cycle time scatter plot

This chart shows the distribution of cycle times across individual tasks. Most tasks were completed within 1–3 days, indicating a relatively stable workflow with no extreme outliers.

Cycle Time Gantt Chart



Cycle time Gantt

The Gantt chart illustrates how tasks were distributed over the sprint timeline. It shows a clear progression from pipeline setup to feature engineering and final documentation, with some parallel work between team members.